Supplementary Material for Searching Identity details across Local-Global Features for Generalized Cross-Domain ECG Recognition

The supplementary material majorly discusses the datasets and experimental protocols used for this work. In Section 3, we report an ablation study on different hyperparameters of the proposed LGTraNet.

1. Datasets

1) MIT-BIH [9]: This is a single-session data collected from a 47 users in ambulatory conditions. The total recording length per subject is 30 minutes. Signal acquisition has been performed using 2 leads placed in chest area. We use the signal from the first lead for experimentation. Each of the training-testing sample extracted from this dataset was composed of 5 R peaks. It must also be noted that certain records had arrhythmia in them.

2) PTB [2]: This dataset comprises of multi-session recordings (average separation between recordings: 23 days) from 290 subjects. Subjects face different forms of cardiac disorder as well. Each recording lasted for 2 minutes. Also, this dataset is collected as per standard 12 lead configuration with additional 3 Frank leads. We use the first lead for experimentation. Each of the extracted signal comprised 2 R peaks.

3) ECG1D [10]: This is also a multi-session dataset, collected from 89 healthy subjects. The recordings lasted for 20 seconds, and were 1 day to 6 months apart. Data has been collected using a single lead placed at wrist. We used 2 R peaks for each of the training and testing sample.

2. Evaluation protocols

We evaluate our method under 4 settings using standard recognition accuracy and equal error rates. Next, we describe the protocols used for each of these settings over the 3 datasets:

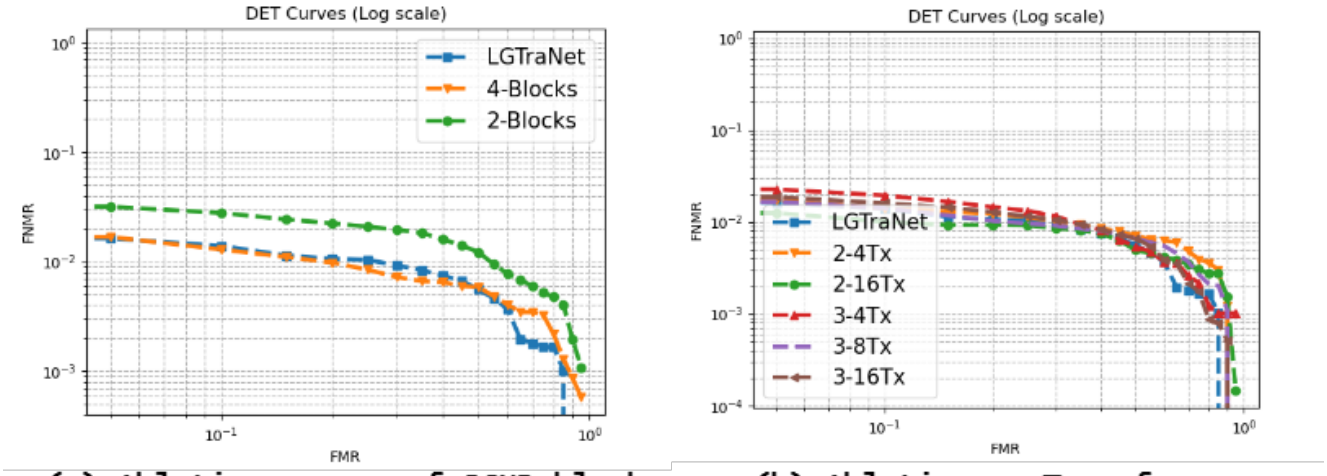
1) Single-session evaluation: In this setting we trained and tested our model using data from single-session. This method of evaluation has been utilized by multiple previous works [1, 7, 13, 5, 12, 14, 8], but because of limited temporal separation, it is not a challenging protocol. This protocol was utilized on following datasets:

1. **MIT-BIH:** Experimentation was performed for all 47 subjects. Initial 50% of each record was used for training, while the remaining for testing.
2. **PTB-290:** All the records from 290 subjects was used. Similar to MIT-BIH, initial 50% of each record was used for training, while the remaining for testing.
3. **PTB-100:** Here, initial 100 subjects from the PTB were selected. Then, each of their recordings was split into 2 halves. Samples extracted from the first and second half were respectively used in training and testing.

2) Multi-session evaluation: In this setting, we train and test our model on recordings from different sessions. Following data settings was utilized for this evaluation:

1. **ECG1D:** Since, ECG1D is a multi-session dataset, we train our model in the first recording while test it on the second. All 89 subjects were included in this experimentation. Also, this evaluation has been advocated by multiple previous works [4, 11, 3, 6, 12]. We conduct ablation study on LGTraNet architecture and incremental learning experiments on this protocol. For pre-training we utilize MIT-BIH dataset. We chose MIT-BIH instead of PTB because PTB is affected by different health conditions. This might lead to learning of biased identity details.
2. **PTB (multi-session):** In this experiment, we selected all those subjects from PTB dataset which had 2 or more recordings. Since, each of the recording corresponds to a different session, we trained our model using the first recording while testing it on the second.

3) Off-the-person evaluation (open-set): In this setting, we consider cross-subject evaluation. We train the model for certain subjects, but evaluate it on others. To be specific, we firstly trained the model on initial 70% subjects. Then, remaining 30% were split into gallery and probe. From the trained model we extracted features for the gallery and probe, and then results for matching experiment is reported. For MIT-BIH dataset AND PTB (290 subjects), we



(a) Ablation on no. of SCNR-blocks (b) Ablation on Transformer

Figure S1. DET plots (log scale) obtained for ablation study on different number of (i) SC-Conv blocks, and (ii) encoder layers and attention heads. In (b), notation is 'A-B Tx', and 'A', 'B' respectively represents number of encoder layers and attention heads.

split gallery and probe as defined in single-session evaluation. However, for ECG1D we respectively utilize training and testing set of multi-session evaluation as gallery and probe. Alongside regular training, we additionally utilize incremental learning for ECG1D. For pre-training we utilize MIT-BIH dataset.

4) Cross-database evaluation: It is essential for a model to generalize across different acquisition settings. Although preprocessing and adequate normalization is performed but data acquisition protocols differ across medical centre. For instance, the number of leads and their placement maybe different. Also, acquisition devices vary centre to centre. To evaluate our proposed model under these scenarios, we perform cross-database evaluation. To be specific, we have two training-testing strategies: (i) we train on one dataset, and then test for the subjects of any other dataset. (ii) We train our model using any two datasets and then evaluate it on the third dataset. The protocol for each of the dataset is as follows:

1. **MIT-BIH:** When training is performed on MIT-BIH, we considered entire MIT-BIH dataset. But when testing is performed on this dataset, we used single-session 50-50 split (as discussed above) as gallery and probe respectively.

2. **PTB:** For training and testing on PTB, we have used the subjects with multiple sessions, i.e., PTB (multi-session). Specifically, during evaluation, we use the first session as the gallery while the second as probe.

3. **ECG1D:** When training is done on ECG1D, we considered recordings from both of the sessions, but for evaluation, we use samples from the first session as gallery and the second as probe.

3. Ablation study on LGTraNet

We have ablation study on number of SCNR-blocks and transformer (refer Figure S1). It is evident from the same that the hyperparameter setting of LGTraNet is optimal and attains the least error rates.

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